
MATH 339 Project Guidelines

Overview

Your group is required to complete a project on one of the topics listed below. If you wish to choose a topic not listed, you must first get approval from your instructor. Follow the guidelines provided to complete your project, which you will present to the class in a 15-20 minute presentation at the end of the semester.

In addition to the presentation, you will need to submit a PowerPoint or Beamer file, along with a reflection on your project. Each group must also create an original mathematical meme related to the project topic. The group with the best meme, as judged by your instructor, will earn bonus points for their project!

Project Topics

Project details can be found here: bit.ly/30IM8gT

Your group can select one of the following project topics from the textbook:

Chapter 1 Projects:

1. *Interpolating Polynomials*: This project shows how to use a system of linear equations to fit a polynomial through a set of points.
2. *Splines*: This project also shows how to use a system of linear equations to fit a piecewise polynomial curve through a set of points.
3. *Network Flows*: The purpose of this project is to show how systems of linear equations may be used to model flow through a network.
4. *The Art of Linear Transformations*: In this project, it is illustrated how to graph a polygon and then use linear transformations to change its shape and create a design.
5. *Loop Currents*: The purpose of this project is to provide more and larger examples of loop currents.
6. *Diet*: The purpose of this project is to provide examples of vector equations that result from balancing nutrients in a diet

Chapter 2 Projects:

7. *Other Matrix Products*: This project introduces two new operations on square matrices called the Jordan product and the commutator product and their properties are explored.
8. *Adjacency Matrices*: The purpose of this project is to show how powers of a matrix may be used to investigate graphs.
9. *Dominance Matrices*: The purpose of this project is to apply matrices and their powers to questions concerning various forms of competition between individuals and groups.
10. *Condition Numbers*: The purpose of this project is to show how a condition number of a matrix A may be defined, and how its value affects the accuracy of solutions to systems of equations $A\vec{x} = \vec{b}$.
11. *Equilibrium Temperature Distributions*: The purpose of this project is to discuss a physical situation in which solving a system of linear equations becomes necessary: that of determining the equilibrium temperature of a thin plate.
12. *The Art of Linear Transformations*: This project illustrates how to graph a polygon and then use linear transformations to move it around in the plane

Chapter 3 Projects:

13. *Weighing Design*: This project develops the concept of weighing design and corresponding matrices for use in weighing a few small, light objects.
14. *Jacobians*: This set of exercises examines how a particular determinant called the Jacobian may be used to allow us to change variables in double and triple integrals.

Chapter 4 Projects:

15. *Exploring Subspaces*: This project explores subspaces with a more hands-on approach.
16. *Hill Substitution Ciphers*: This project shows how to use matrices to encode and decode messages.
17. *Error Detecting and Error Correcting*: In this project, a method detecting and correcting errors made in the transmission of encoded messages is constructed. It will turn out that abstract vector spaces and the concepts of null space, rank, and dimension are needed for this construction.
18. *Signal Processing*: This project examines signal processing in more detail.
19. *Fibonacci Sequences*: The purpose of this project is to investigate further the Fibonacci sequence, which arises in number theory, applied mathematics, and biology

Chapter 5 Projects:

20. *Power Method for Finding Eigenvalues*: This project shows how to find the eigenvector associated with the eigenvalue corresponding to the largest eigenvalue.
21. *Integration by Parts*: The purpose of this project is to show how the matrix of a linear transformation relative to a basis may be used to find antiderivatives usually found using integration by parts.
22. *Robotics*: In this project, students are asked to find online examples of robots that use 3D rotations to function.

Project Components

The project is worth a total of 40 points distributed like so:

Presentation (10 points)

- Prepare a 15–20 minute presentation to share your project with the class at the end of the semester.
- Every group member must contribute meaningfully to the presentation.
- Your presentation should include:
 - Mathematical background and key ideas of your topic
 - Examples or visualizations to help your classmates understand the concept
 - Applications of your topic to real-world problems, technology, or other fields of mathematics
- Each student will be assessed individually on oral communication using the rubric posted on Blackboard.
- The group's overall presentation grade will be the average of all individual presentation scores.
- Please upload your presentation slides (PowerPoint, Beamer file, or shareable link) as a group before class on your presentation day.

Presentation Slides (5 points)

- Upload your PowerPoint or Beamer file (or a link) on Blackboard.
- Include the following:
 - Meme: Place your mathematical meme on the next-to-last slide.

- References: Add a list of sources on the final slide.

Mathematical Meme (5 points)

- Upload a high-quality image of your original meme related to your project topic.
- If you've created multiple memes for fun, select one to be graded and entered into the "Best Meme" competition for bonus points.
- Your meme must clearly relate to your project topic to earn full credit.
- No profanity or inappropriate content. Violation will result in 0 points for the meme component.

Reflection (10 points)

Write and upload a one-page, single-spaced reflection (12-point font) addressing the following:

- What is the most valuable thing you learned from this project?
- How did the project deepen your understanding of your topic or related concepts?
- What skills (e.g., technical, problem-solving, communication) did you improve or develop?
- How do you think these skills will benefit you in future courses or your career?
- In one sentence, summarize your biggest takeaway from this project.

Individual Participation Rubric (10 points)

- Complete the peer evaluation rubric to assess your group members' participation.
- Your individual participation grade will be calculated as the average of the scores given by your group members.
- You do not grade yourself—only your peers.
- Failure to submit a completed rubric will result in a 0 for your participation grade, regardless of the scores your peers assign.